

ILLUSTRATIVE EXAMPLES**LEVEL-1****Type I ON FINDING THE GENERAL SOLUTIONS OF THE EQUATIONS OF THE FORM**

$$\sin \theta = \sin \alpha, \cos \theta = \cos \alpha, \tan \theta = \tan \alpha$$

EXAMPLE 1 Find the general solutions of the following equations:

$$(i) \sin \theta = \frac{\sqrt{3}}{2} \quad (ii) 2 \sin \theta + 1 = 0 \quad (iii) \operatorname{cosec} \theta = 2$$

SOLUTION (i) A value of θ satisfying $\sin \theta = \frac{\sqrt{3}}{2}$ is $\frac{\pi}{3}$.

$$\therefore \sin \theta = \frac{\sqrt{3}}{2} \Rightarrow \sin \theta = \sin \frac{\pi}{3} \Rightarrow \theta = n\pi + (-1)^n \frac{\pi}{3}, n \in \mathbb{Z}$$

(ii) We have,

$$2 \sin \theta + 1 = 0 \Rightarrow \sin \theta = -\frac{1}{2}$$

A value of θ satisfying this equation is $-\pi/6$.

$$\therefore \sin \theta = -\frac{1}{2}$$

$$\Rightarrow \sin \theta = \sin \left(-\frac{\pi}{6} \right)$$

$$\Rightarrow \theta = n\pi + (-1)^n \left(-\frac{\pi}{6} \right), n \in \mathbb{Z}$$

$$\Rightarrow \theta = n\pi + (-1)^{n+1} \frac{\pi}{6}, n \in \mathbb{Z}.$$

(iii) We have,

$$\operatorname{cosec} \theta = 2 \Rightarrow \sin \theta = \frac{1}{2} \Rightarrow \sin \theta = \sin \frac{\pi}{6} \Rightarrow \theta = n\pi + (-1)^n \frac{\pi}{6}, n \in \mathbb{Z}.$$

EXAMPLE 2 Find the general solutions of the following equations:

$$(i) \cos \theta = \frac{1}{2} \quad (ii) \cos 3\theta = -\frac{1}{2} \quad (iii) \sqrt{3} \sec 2\theta = 2$$

SOLUTION (i) $\cos \theta = \frac{1}{2} \Rightarrow \cos \theta = \cos \frac{\pi}{3} \Rightarrow \theta = 2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$

$$(ii) \cos 3\theta = -\frac{1}{2}$$

$$\Rightarrow \cos 3\theta = \cos \frac{2\pi}{3}$$

$$\Rightarrow 3\theta = 2n\pi \pm \frac{2\pi}{3}, n \in \mathbb{Z}$$

$$\Rightarrow \theta = \frac{2n\pi}{3} \pm \frac{2\pi}{9}, n \in \mathbb{Z}$$

$$(iii) \sqrt{3} \sec 2\theta = 2$$

$$\Rightarrow \cos 2\theta = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \cos 2\theta = \cos \frac{\pi}{6}$$

$$\Rightarrow 2\theta = 2n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$$

$$\Rightarrow \theta = n\pi \pm \frac{\pi}{12}, n \in \mathbb{Z}$$

EXAMPLE 3 Solve the following trigonometric equations:

$$(i) \tan \theta = \frac{1}{\sqrt{3}}$$

$$(ii) \tan 2\theta = \sqrt{3}$$

$$(iii) \tan 3\theta = -1$$

SOLUTION (i) $\tan \theta = \frac{1}{\sqrt{3}} \Rightarrow \tan \theta = \tan \frac{\pi}{6} \Rightarrow \theta = n\pi + \frac{\pi}{6}, n \in \mathbb{Z}$

$$(ii) \quad \tan 2\theta = \sqrt{3}$$

$$\Rightarrow \tan 2\theta = \tan \frac{\pi}{3}$$

$$\Rightarrow 2\theta = n\pi + \frac{\pi}{3}, n \in \mathbb{Z}$$

$$\Rightarrow \theta = \frac{n\pi}{2} + \frac{\pi}{6}, n \in \mathbb{Z}$$

$$(iii) \quad \tan 3\theta = -1$$

$$\Rightarrow \tan 3\theta = \tan \left(-\frac{\pi}{4}\right)$$

$$\Rightarrow 3\theta = n\pi + \left(-\frac{\pi}{4}\right), n \in \mathbb{Z}$$

$$\Rightarrow \theta = \frac{n\pi}{3} - \frac{\pi}{12}, n \in \mathbb{Z}.$$

EXAMPLE 4 Solve the following trigonometric equations:

$$(i) \sin \frac{\theta}{2} = -1$$

$$(ii) \cos \frac{3\theta}{2} = \frac{1}{2}$$

$$(iii) \tan \left(\frac{2}{3}\theta\right) = \sqrt{3}$$

SOLUTION (i) $\sin \frac{\theta}{2} = -1$

$$\Rightarrow \sin \frac{\theta}{2} = \sin \left(-\frac{\pi}{2}\right)$$

$$\Rightarrow \frac{\theta}{2} = n\pi + (-1)^n \left(-\frac{\pi}{2}\right), n \in \mathbb{Z}$$

$$\Rightarrow \theta = 2n\pi + (-1)^{n+1} \pi, n \in \mathbb{Z}$$

$$(ii) \quad \cos \frac{3\theta}{2} = \frac{1}{2}$$

$$\Rightarrow \cos \frac{3\theta}{2} = \cos \frac{\pi}{3}$$

$$\Rightarrow \frac{3\theta}{2} = 2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$$

$$\Rightarrow \theta = \frac{4n\pi}{3} \pm \frac{2\pi}{9}, n \in \mathbb{Z}$$

$$(iii) \quad \tan \left(\frac{2\theta}{3}\right) = \sqrt{3}$$

$$\Rightarrow \tan \left(\frac{2\theta}{3}\right) = \tan \frac{\pi}{3}$$

$$\Rightarrow \frac{2\theta}{3} = n\pi + \frac{\pi}{3}, n \in \mathbb{Z}$$

$$\Rightarrow \theta = \frac{3n\pi}{2} + \frac{\pi}{2}, n \in \mathbb{Z}.$$

Type II ON FINDING THE GENERAL SOLUTION OF THE EQUATIONS REDUCIBLE TO THE FORMS

$$\sin \theta = \sin \alpha, \cos \theta = \cos \alpha, \tan \theta = \tan \alpha.$$

EXAMPLE 5 Solve the equation: $\sin \theta + \sin 3\theta + \sin 5\theta = 0$.**[NCERT EXEMPLAR]****SOLUTION** We have,

$$\sin \theta + \sin 3\theta + \sin 5\theta = 0$$

$$\Rightarrow (\sin 5\theta + \sin \theta) + \sin 3\theta = 0$$

$$\Rightarrow 2 \sin 3\theta \cos 2\theta + \sin 3\theta = 0$$

$$\Rightarrow \sin 3\theta (2 \cos 2\theta + 1) = 0$$

$$\Rightarrow \sin 3\theta = 0 \text{ or } 2 \cos 2\theta + 1 = 0$$

$$\Rightarrow \sin 3\theta = 0 \text{ or, } \cos 2\theta = -\frac{1}{2}$$

$$\text{Now, } \sin 3\theta = 0 \Rightarrow 3\theta = n\pi, n \in \mathbb{Z} \Rightarrow \theta = \frac{n\pi}{3}, n \in \mathbb{Z}$$

$$\text{And, } \cos 2\theta = -\frac{1}{2}$$

$$\Rightarrow \cos 2\theta = \cos \frac{2\pi}{3}$$

$$\Rightarrow 2\theta = 2m\pi \pm \frac{2\pi}{3}, m \in \mathbb{Z}$$

$$\Rightarrow \theta = m\pi \pm \frac{\pi}{3}, m \in \mathbb{Z}.$$

Hence, the general solution of the given equation is: $\theta = \frac{n\pi}{3}$ or, $\theta = m\pi \pm \frac{\pi}{3}$, where $m, n \in \mathbb{Z}$.

EXAMPLE 6 Solve the equation: $\cos \theta + \cos 3\theta - 2 \cos 2\theta = 0$ **SOLUTION** We have,

$$\cos \theta + \cos 3\theta - 2 \cos 2\theta = 0$$

$$\Leftrightarrow 2 \cos 2\theta \cos \theta - 2 \cos 2\theta = 0$$

$$\Leftrightarrow 2 \cos 2\theta (\cos \theta - 1) = 0$$

$$\Rightarrow \cos 2\theta = 0 \text{ or, } \cos \theta - 1 = 0$$

$$\text{Now, } \cos 2\theta = 0 \Rightarrow 2\theta = (2n+1) \frac{\pi}{2}, n \in \mathbb{Z} \Rightarrow \theta = (2n+1) \frac{\pi}{4}, n \in \mathbb{Z}$$

$$\text{And, } \cos \theta - 1 = 0$$

$$\Rightarrow \cos \theta = 1$$

$$\Rightarrow \cos \theta = \cos 0$$

$$\Rightarrow \theta = 2m\pi \pm 0, m \in \mathbb{Z}$$

$$\Rightarrow \theta = 2m\pi, m \in \mathbb{Z}$$

Hence, $\theta = (2n+1) \frac{\pi}{4}$ or, $\theta = 2m\pi$, where $m, n \in \mathbb{Z}$.

EXAMPLE 7 Solve the equation: $\sin m\theta + \sin n\theta = 0$.**SOLUTION** We have,

$$\sin m\theta + \sin n\theta = 0$$

$$\Rightarrow 2 \sin \left(\frac{m+n}{2} \theta \right) \cos \left(\frac{m-n}{2} \theta \right) = 0$$

$$\Rightarrow \sin \left(\frac{m+n}{2} \theta \right) = 0 \text{ or, } \cos \left(\frac{m-n}{2} \theta \right) = 0$$

Now, $\sin\left(\frac{m+n}{2}\right)\theta = 0$

$$\Rightarrow \left(\frac{m+n}{2}\right)\theta = r\pi, r \in \mathbb{Z}$$

$$\Rightarrow \theta = \frac{2r\pi}{m+n}, r \in \mathbb{Z}$$

And, $\cos\left(\frac{m-n}{2}\right)\theta = 0$

$$\Rightarrow \left(\frac{m-n}{2}\right)\theta = (2s+1)\frac{\pi}{2}, s \in \mathbb{Z}$$

$$\Rightarrow \theta = \frac{(2s+1)\pi}{m-n}, s \in \mathbb{Z}$$

Hence, $\theta = \frac{2r\pi}{m+n}$ or, $\theta = \frac{(2s+1)\pi}{m-n}$, where $r, s \in \mathbb{Z}$.

EXAMPLE 8 Solve the following equations:

(i) $\sin 2\theta + \cos \theta = 0$ [NCERT]

(ii) $\sin 3\theta + \cos 2\theta = 0$

(iii) $\sin 2\theta + \sin 4\theta + \sin 6\theta = 0$

SOLUTION (i) $\sin 2\theta + \cos \theta = 0$

$$\Rightarrow \cos \theta = -\sin 2\theta$$

$$\Rightarrow \cos \theta = \cos\left(\frac{\pi}{2} + 2\theta\right)$$

$$\Rightarrow \theta = 2n\pi \pm \left(\frac{\pi}{2} + 2\theta\right), n \in \mathbb{Z}$$

Taking positive sign, we have

$$\theta = 2n\pi + \frac{\pi}{2} + 2\theta$$

$$\Rightarrow -\theta = 2n\pi + \frac{\pi}{2}, n \in \mathbb{Z}$$

$$\Rightarrow \theta = -2n\pi - \frac{\pi}{2}, n \in \mathbb{Z}$$

$$\Rightarrow \theta = 2m\pi - \frac{\pi}{2}, \text{ where } m = -n \in \mathbb{Z}.$$

Taking negative sign, we have

$$\theta = 2n\pi - \left(\frac{\pi}{2} + 2\theta\right)$$

$$\Rightarrow 3\theta = 2n\pi - \frac{\pi}{2} \Rightarrow \theta = \frac{2n\pi}{3} - \frac{\pi}{6}, n \in \mathbb{Z}.$$

Hence, $\theta = 2m\pi - \frac{\pi}{2}$, or, $\theta = \frac{2n\pi}{3} - \frac{\pi}{6}$, where $m, n \in \mathbb{Z}$.

(ii) $\sin 3\theta + \cos 2\theta = 0$

$$\Rightarrow \cos 2\theta = -\sin 3\theta$$

$$\Rightarrow \cos 2\theta = \cos\left(\frac{\pi}{2} + 3\theta\right)$$

$$\Rightarrow 2\theta = 2n\pi \pm \left(\frac{\pi}{2} + 3\theta\right), n \in \mathbb{Z}$$

Taking positive sign, we have

$$2\theta = 2n\pi + \frac{\pi}{2} + 3\theta$$

$$\Rightarrow -\theta = 2n\pi + \frac{\pi}{2}$$

$$\Rightarrow \theta = -2n\pi - \frac{\pi}{2}$$

$$\Rightarrow \theta = 2m\pi - \frac{\pi}{2}, \text{ where } -n = m.$$

Taking negative sign, we have

$$2\theta = 2n\pi - \frac{\pi}{2} - 3\theta \Rightarrow 5\theta = 2n\pi - \frac{\pi}{2} \Rightarrow \theta = \frac{2n\pi}{5} - \frac{\pi}{10}, n \in \mathbb{Z}$$

Hence, $\theta = \frac{2n\pi}{5} - \frac{\pi}{10}$ or, $\theta = 2m\pi - \frac{\pi}{2}$, where $m, n \in \mathbb{Z}$.

(iii) We have,

$$\sin 2\theta + \sin 4\theta + \sin 6\theta = 0$$

$$\Rightarrow \sin 4\theta + (\sin 2\theta + \sin 6\theta) = 0$$

$$\Rightarrow \sin 4\theta + 2 \sin 4\theta \cos 2\theta = 0$$

$$\Rightarrow \sin 4\theta (1 + 2 \cos 2\theta) = 0$$

$$\Rightarrow \sin 4\theta = 0 \text{ or, } 1 + 2 \cos 2\theta = 0 \Rightarrow \sin 4\theta = 0 \text{ or, } \cos 2\theta = -\frac{1}{2}$$

Now, $\sin 4\theta = 0 \Rightarrow 4\theta = n\pi, n \in \mathbb{Z} \Rightarrow \theta = \frac{n\pi}{4}, n \in \mathbb{Z}$

And, $\cos 2\theta = -\frac{1}{2}$

$$\Rightarrow \cos 2\theta = \cos \frac{2\pi}{3}$$

$$\Rightarrow 2\theta = 2m\pi \pm \frac{2\pi}{3}, m \in \mathbb{Z}$$

$$\Rightarrow \theta = m\pi \pm \frac{\pi}{3}, m \in \mathbb{Z}$$

Hence, $\theta = \frac{n\pi}{4}$ or, $\theta = m\pi \pm \frac{\pi}{3}$, where $m, n \in \mathbb{Z}$.

EXAMPLE 9 Solve the following equations:

(i) $2 \cos^2 \theta + 3 \sin \theta = 0$ [NCERT] (ii) $\cot^2 \theta + \frac{3}{\sin \theta} + 3 = 0$

(iii) $2 \tan \theta - \cot \theta = -1$

(iv) $4 \cos \theta - 3 \sec \theta = \tan \theta$

(v) $\tan^2 \theta + (1 - \sqrt{3}) \tan \theta - \sqrt{3} = 0$ (vi) $\sec^2 2x = 1 - \tan 2x$

[NCERT]

SOLUTION (i) $2 \cos^2 \theta + 3 \sin \theta = 0$

$$\Rightarrow 2(1 - \sin^2 \theta) + 3 \sin \theta = 0$$

$$\Rightarrow 2 \sin^2 \theta - 3 \sin \theta - 2 = 0$$

$$\Rightarrow 2 \sin^2 \theta - 4 \sin \theta + \sin \theta - 2 = 0$$

$$\Rightarrow 2 \sin \theta (\sin \theta - 2) + 1 (\sin \theta - 2) = 0$$

$$\Rightarrow (\sin \theta - 2) (2 \sin \theta + 1) = 0$$

$$\Rightarrow 2 \sin \theta + 1 = 0$$

$$[\because \sin \theta \neq 2 \therefore \sin \theta - 2 \neq 0]$$

$$\Rightarrow \sin \theta = -\frac{1}{2}$$

$$\Rightarrow \sin \theta = \sin \left(-\frac{\pi}{6} \right)$$

$$\Rightarrow \theta = n\pi + (-1)^n \left(-\frac{\pi}{6} \right), n \in \mathbb{Z}$$

$$\Rightarrow \theta = n\pi + (-1)^{n+1} \frac{\pi}{6}, n \in \mathbb{Z}.$$

$$(ii) \cot^2 \theta + \frac{3}{\sin \theta} + 3$$

$$\Rightarrow \operatorname{cosec}^2 \theta$$

$$\Rightarrow \cot^2 \theta$$

$$\Rightarrow$$

$$\Rightarrow$$

Now,

$$\Rightarrow$$

$$\Rightarrow \sin$$

$$\Rightarrow \sin \theta = \sin \left(-\frac{\pi}{6} \right)$$

$$\Rightarrow \theta = n\pi + (-1)^n \left(-\frac{\pi}{6} \right), n \in \mathbb{Z}$$

$$\Rightarrow \theta = n\pi + (-1)^{n+1} \frac{\pi}{6}, n \in \mathbb{Z}$$

$$\text{And, } \operatorname{cosec} \theta + 1 = 0$$

$$\Rightarrow \frac{1}{\sin \theta} + 1 = 0$$

$$\Rightarrow \sin \theta = -1$$

$$\Rightarrow \sin \theta = \sin \left(-\frac{\pi}{2} \right)$$

$$\Rightarrow \theta = m\pi + (-1)^m \left(-\frac{\pi}{2} \right), m \in \mathbb{Z}$$

$$\Rightarrow \theta = m\pi + (-1)^{m+1} \frac{\pi}{2}, m \in \mathbb{Z}$$

$$\text{Hence, } \theta = n\pi + (-1)^{n+1} \frac{\pi}{6} \text{ or, } \theta = m\pi + (-1)^{m+1} \frac{\pi}{2}, m, n \in \mathbb{Z}$$

$$(iii) 2 \tan \theta - \cot \theta = -1$$

$$\Rightarrow 2 \tan \theta - \frac{1}{\tan \theta} = -1$$

$$\Rightarrow 2 \tan^2 \theta + \tan \theta - 1 = 0$$

$$\Rightarrow 2 \tan^2 \theta + 2 \tan \theta - \tan \theta - 1 = 0$$

$$\Rightarrow 2 \tan \theta (\tan \theta + 1) - (\tan \theta + 1) = 0$$

$$\Rightarrow (\tan \theta + 1) (2 \tan \theta - 1) = 0$$

$$\Rightarrow \tan \theta = -1 \text{ or, } \tan \theta = \frac{1}{2}$$

$$\text{Now, } \tan \theta = -1$$

$$\Rightarrow \tan \theta = \tan \left(-\frac{\pi}{4} \right)$$

$$\Rightarrow \theta = n\pi + \left(-\frac{\pi}{4} \right), n \in \mathbb{Z}$$

$$\Rightarrow \theta = n\pi - \frac{\pi}{4}, n \in \mathbb{Z}$$

$$\text{And, } \tan \theta = \frac{1}{2}$$

$$\Rightarrow \tan \theta = \tan \alpha, \text{ where } \tan \alpha = \frac{1}{2}$$

$$\Rightarrow \theta = m\pi + \alpha, \text{ where } \tan \alpha = \frac{1}{2} \text{ and } m \in \mathbb{Z}$$

$$\text{Hence, } \theta = n\pi - \frac{\pi}{4} \text{ or, } \theta = m\pi + \alpha, \text{ where } m, n \in \mathbb{Z} \text{ and } \tan \alpha = \frac{1}{2}$$

$$(iv) \quad 4 \cos \theta - 3 \sec \theta = \tan \theta$$

$$\Rightarrow 4 \cos \theta - \frac{3}{\cos \theta} = \frac{\sin \theta}{\cos \theta}$$

$$\Rightarrow 4 \cos^2 \theta - 3 = \sin \theta$$

$$\Rightarrow 4(1 - \sin^2 \theta) - 3 = \sin \theta$$

$$\Rightarrow 4 \sin^2 \theta + \sin \theta - 1 = 0$$

$$\Rightarrow \sin \theta = \frac{-1 \pm \sqrt{1+16}}{8}$$

$$\Rightarrow \sin \theta = \frac{-1 \pm \sqrt{17}}{8}$$

$$\Rightarrow \sin \theta = \frac{-1 + \sqrt{17}}{8} \text{ or, } \sin \theta = \frac{-1 - \sqrt{17}}{8}$$

$$\text{Now, } \sin \theta = \frac{-1 + \sqrt{17}}{8}$$

$$\Rightarrow \sin \theta = \sin \alpha, \text{ where } \sin \alpha = \frac{-1 + \sqrt{17}}{8}$$

$$\Rightarrow \theta = n\pi + (-1)^n \alpha, \text{ where } \sin \alpha = \frac{-1 + \sqrt{17}}{8} \text{ and } n \in \mathbb{Z}$$

$$\text{And, } \sin \theta = \frac{-1 - \sqrt{17}}{8}$$

$$\Rightarrow \sin \theta = \sin \beta, \text{ where } \sin \beta = \frac{-1 - \sqrt{17}}{8}$$

$$\Rightarrow \theta = n\pi + (-1)^n \beta, \text{ where } \sin \beta = \frac{-1 - \sqrt{17}}{8}$$

$$(v) \quad \tan^2 \theta + (1 - \sqrt{3}) \tan \theta - \sqrt{3} = 0$$

$$\Rightarrow \tan^2 \theta + \tan \theta - \sqrt{3} \tan \theta - \sqrt{3} = 0$$

$$\Rightarrow \tan (\tan \theta + 1) - \sqrt{3} (\tan \theta + 1) = 0$$

$$\Rightarrow (\tan \theta + 1) (\tan \theta - \sqrt{3}) = 0$$

$$\Rightarrow \tan \theta + 1 = 0 \text{ or } \tan \theta - \sqrt{3} = 0$$

$$\Rightarrow \tan \theta = -1 \text{ or, } \tan \theta = \sqrt{3}$$

$$\text{Now, } \tan \theta = -1 \Rightarrow \tan \theta = \tan \left(-\frac{\pi}{4} \right) \Rightarrow \theta = n\pi - \frac{\pi}{4}, n \in \mathbb{Z}$$

$$\text{And, } \tan \theta = \sqrt{3} \Rightarrow \tan \theta = \tan \frac{\pi}{3} \Rightarrow \theta = m\pi + \frac{\pi}{3}, m \in \mathbb{Z}$$

$$\text{Hence, } \theta = n\pi - \frac{\pi}{4} \text{ or, } \theta = m\pi + \frac{\pi}{3}, \text{ where } m, n \in \mathbb{Z}.$$

$$(vi) \sec^2 2x = 1 - \tan 2x$$

$$\Rightarrow 1 + \tan^2 2x = 1 - \tan 2x$$

$$\Rightarrow \tan^2 2x + \tan 2x = 0$$

$$\Rightarrow \tan 2x (\tan 2x + 1) = 0$$

$$\Rightarrow \tan 2x = 0 \text{ or, } \tan 2x + 1 = 0$$

$$\Rightarrow \tan 2x = 0 \text{ or, } \tan 2x = -1$$

$$\Rightarrow \tan 2x = 0 \text{ or, } \tan 2x = \tan \frac{3\pi}{4}$$

$$\Rightarrow 2x = n\pi \text{ or, } 2x = n\pi + \frac{3\pi}{4}, n \in \mathbb{Z}$$

$$\Rightarrow x = \frac{n\pi}{2} \text{ or, } x = \frac{n\pi}{2} + \frac{3\pi}{8}, n \in \mathbb{Z}$$

EXAMPLE 10 Solve the following equations:

$$(i) \tan \theta + \tan 2\theta + \tan \theta \tan 2\theta = 1$$

$$(ii) \tan \theta + \tan 2\theta + \tan 3\theta = \tan \theta \tan 2\theta \tan 3\theta$$

$$(iii) \tan \theta + \tan 2\theta + \sqrt{3} \tan \theta \tan 2\theta = \sqrt{3} \quad (iv) \tan \theta + \tan \left(\theta + \frac{\pi}{3} \right) + \tan \left(\theta + \frac{2\pi}{3} \right) = 3$$

SOLUTION (i) $\tan \theta + \tan 2\theta + \tan \theta \tan 2\theta = 1$

$$\Rightarrow \tan \theta + \tan 2\theta = 1 - \tan \theta \tan 2\theta$$

$$\Rightarrow \frac{\tan \theta + \tan 2\theta}{1 - \tan \theta \tan 2\theta} = 1$$

$$\Rightarrow \tan 3\theta = 1$$

$$\Rightarrow \tan 3\theta = \tan \frac{\pi}{4}$$

$$\Rightarrow 3\theta = n\pi + \frac{\pi}{4}, n \in \mathbb{Z}$$

$$\Rightarrow \theta = \frac{n\pi}{3} + \frac{\pi}{12}, n \in \mathbb{Z}$$

$$(ii) \tan \theta + \tan 2\theta + \tan 3\theta = \tan \theta \tan 2\theta \tan 3\theta$$

$$\Rightarrow \tan \theta + \tan 2\theta = -\tan 3\theta + \tan \theta \tan 2\theta \tan 3\theta$$

$$\Rightarrow \tan \theta + \tan 2\theta = -\tan 3\theta (1 - \tan \theta \tan 2\theta)$$

$$\Rightarrow \frac{\tan \theta + \tan 2\theta}{1 - \tan \theta \tan 2\theta} = -\tan 3\theta$$

$$\Rightarrow \tan (\theta + 2\theta) = -\tan 3\theta$$

$$\Rightarrow \tan 3\theta = -\tan 3\theta$$

$$\Rightarrow 2\tan 3\theta = 0$$

$$\Rightarrow \tan 3\theta = 0$$

$$\Rightarrow 3\theta = n\pi, n \in \mathbb{Z} \Rightarrow \theta = \frac{n\pi}{3}, n \in \mathbb{Z}.$$

$$\begin{aligned}
 \text{(iii)} \quad & \tan \theta + \tan 2\theta + \sqrt{3} \tan \theta \tan 2\theta = \sqrt{3} \\
 \Rightarrow \quad & \tan \theta + \tan 2\theta = \sqrt{3} (1 - \tan \theta \tan 2\theta) \\
 \Rightarrow \quad & \frac{\tan \theta + \tan 2\theta}{1 - \tan \theta \tan 2\theta} = \sqrt{3} \\
 \Rightarrow \quad & \tan 3\theta = \sqrt{3} \\
 \Rightarrow \quad & \tan 3\theta = \tan \frac{\pi}{3} \\
 \Rightarrow \quad & 3\theta = n\pi + \frac{\pi}{3}, n \in \mathbb{Z} \\
 \Rightarrow \quad & \theta = \frac{n\pi}{3} + \frac{\pi}{9}, n \in \mathbb{Z}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv)} \quad & \tan \theta + \tan \left(\theta + \frac{\pi}{3} \right) + \tan \left(\theta + \frac{2\pi}{3} \right) = 3 \\
 \Rightarrow \quad & \tan \theta + \frac{\tan \theta + \tan \frac{\pi}{3}}{1 - \tan \theta \tan \frac{\pi}{3}} + \frac{\tan \theta + \tan \frac{2\pi}{3}}{1 - \tan \theta \tan \frac{2\pi}{3}} = 3 \\
 \Rightarrow \quad & \tan \theta + \frac{\tan \theta + \sqrt{3}}{1 - \sqrt{3} \tan \theta} + \frac{\tan \theta - \sqrt{3}}{1 + \sqrt{3} \tan \theta} = 3 \\
 \Rightarrow \quad & \tan \theta + \frac{8 \tan \theta}{1 - 3 \tan^2 \theta} = 3 \\
 \Rightarrow \quad & \frac{\tan \theta - 3 \tan^3 \theta + 8 \tan \theta}{1 - 3 \tan^2 \theta} = 3 \\
 \Rightarrow \quad & \frac{3(3 \tan \theta - \tan^3 \theta)}{1 - 3 \tan^2 \theta} = 3 \\
 \Rightarrow \quad & 3 \tan 3\theta = 3 \\
 \Rightarrow \quad & \tan 3\theta = 1 \\
 \Rightarrow \quad & \tan 3\theta = \tan \frac{\pi}{4} \\
 \Rightarrow \quad & 3\theta = n\pi + \frac{\pi}{4}, n \in \mathbb{Z} \\
 \Rightarrow \quad & \theta = \frac{n\pi}{3} + \frac{\pi}{12}, n \in \mathbb{Z}.
 \end{aligned}$$

11.3 GENERAL SOLUTIONS OF TRIGONOMETRICAL EQUATIONS OF THE FORM

$$\sin^2 \theta = \sin^2 \alpha, \cos^2 \theta = \cos^2 \alpha, \tan^2 \theta = \tan^2 \alpha$$

THEOREM Prove that:

$$\begin{aligned}
 \text{(i)} \quad & \sin^2 \theta = \sin^2 \alpha \Rightarrow \theta = n\pi \pm \alpha, n \in \mathbb{Z} \\
 \text{(ii)} \quad & \cos^2 \theta = \cos^2 \alpha \Rightarrow \theta = n\pi \pm \alpha, n \in \mathbb{Z} \\
 \text{(iii)} \quad & \tan^2 \theta = \tan^2 \alpha \Rightarrow \theta = n\pi \pm \alpha, n \in \mathbb{Z}
 \end{aligned}$$

PROOF (i) $\sin^2 \theta = \sin^2 \alpha$

$$\begin{aligned}
 \Rightarrow \quad & 2 \sin^2 \theta = 2 \sin^2 \alpha \\
 \Rightarrow \quad & 1 - \cos 2\theta = 1 - \cos 2\alpha \\
 \Rightarrow \quad & \cos 2\theta = \cos 2\alpha \\
 \Rightarrow \quad & 2\theta = 2n\pi \pm 2\alpha, n \in \mathbb{Z} \\
 \Rightarrow \quad & \theta = n\pi \pm \alpha, n \in \mathbb{Z}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad & \cos^2 \theta = \cos^2 \alpha \\
 \Rightarrow & 2 \cos^2 \theta = 2 \cos^2 \alpha \\
 \Rightarrow & 1 + \cos 2\theta = 1 + \cos 2\alpha \\
 \Rightarrow & \cos 2\theta = \cos 2\alpha \\
 \Rightarrow & 2\theta = 2n\pi \pm 2\alpha, n \in \mathbb{Z} \\
 \Rightarrow & \theta = n\pi \pm \alpha, n \in \mathbb{Z}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad & \tan^2 \theta = \tan^2 \alpha \\
 \Rightarrow & \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \frac{1 - \tan^2 \alpha}{1 + \tan^2 \alpha} \\
 \Rightarrow & \cos 2\theta = \cos 2\alpha \\
 \Rightarrow & 2\theta = 2n\pi \pm 2\alpha, n \in \mathbb{Z} \\
 \Rightarrow & \theta = n\pi \pm \alpha, n \in \mathbb{Z}
 \end{aligned}$$

Q.E.D.

ILLUSTRATIVE EXAMPLES**LEVEL-1****EXAMPLE 1** Solve: $7 \cos^2 \theta + 3 \sin^2 \theta = 4$ **SOLUTION** We have,

$$\begin{aligned}
 & 7 \cos^2 \theta + 3 \sin^2 \theta = 4 \\
 \Rightarrow & 7(1 - \sin^2 \theta) + 3 \sin^2 \theta = 4 \\
 \Rightarrow & 4 \sin^2 \theta = 3 \\
 \Rightarrow & 4 \sin^2 \theta = 3 \\
 \Rightarrow & \sin^2 \theta = \frac{3}{4} = \left(\frac{\sqrt{3}}{2}\right)^2 \\
 \Rightarrow & \sin^2 \theta = \sin^2 \frac{\pi}{3} \Rightarrow \theta = n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}
 \end{aligned}$$

EXAMPLE 2 Solve: $2 \sin^2 x + \sin^2 2x = 2$ **SOLUTION** We have,

$$\begin{aligned}
 & 2 \sin^2 x + \sin^2 2x = 2 \\
 \Rightarrow & 2 \sin^2 x + (2 \sin x \cos x)^2 = 2 \\
 \Rightarrow & 4 \sin^2 x \cos^2 x + 2 \sin^2 x = 2 \\
 \Rightarrow & 2 \sin^2 x \cos^2 x + \sin^2 x = 1 \\
 \Rightarrow & 2 \sin^2 x \cos^2 x - (1 - \sin^2 x) = 0 \\
 \Rightarrow & 2 \sin^2 x \cos^2 x - \cos^2 x = 0 \\
 \Rightarrow & \cos^2 x (2 \sin^2 x - 1) = 0 \Rightarrow \cos^2 x = 0 \text{ or } 2 \sin^2 x - 1 = 0 \\
 \Rightarrow & \cos^2 x = 0 \text{ or, } \sin^2 x = \frac{1}{2}
 \end{aligned}$$

$$\text{Now, } \cos^2 x = 0 \Rightarrow \cos^2 x = \cos^2 \frac{\pi}{2} \Rightarrow x = n\pi \pm \frac{\pi}{2}, n \in \mathbb{Z}$$

$$\text{And, } \sin^2 x = \frac{1}{2} \Rightarrow \sin^2 x = \sin^2 \frac{\pi}{4} \Rightarrow x = m\pi \pm \frac{\pi}{4}, m \in \mathbb{Z}$$

$$\text{Hence, } x = n\pi \pm \frac{\pi}{2} \text{ or } x = m\pi \pm \frac{\pi}{4}, \text{ where } m, n \in \mathbb{Z}$$

LEVEL-2

EXAMPLE 3 Solve: $\sin 3\alpha = 4 \sin \alpha \sin (x + \alpha) \sin (x - \alpha)$, where $\alpha \neq n\pi, n \in \mathbb{Z}$

SOLUTION We have,

$$\begin{aligned} \sin 3\alpha &= 4 \sin \alpha \sin (x + \alpha) \sin (x - \alpha) \\ \Rightarrow \sin 3\alpha &= 4 \sin \alpha (\sin^2 x - \sin^2 \alpha) \\ \Rightarrow 3 \sin \alpha - 4 \sin^3 \alpha &= 4 \sin^2 x \sin \alpha - 4 \sin^3 \alpha \\ \Rightarrow 3 \sin \alpha &= 4 \sin^2 x \sin \alpha \\ \Rightarrow \sin^2 x &= \frac{3}{4} = \left(\frac{\sqrt{3}}{2}\right)^2 \\ \Rightarrow \sin^2 x &= \sin^2 \frac{\pi}{3} \Rightarrow x = n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z} \end{aligned}$$

EXAMPLE 4 Solve: $4 \sin x \sin 2x \sin 4x = \sin 3x$

SOLUTION We have,

$$\begin{aligned} 4 \sin x \sin 2x \sin 4x &= \sin 3x \\ \Rightarrow 4 \sin x \sin (3x - x) \cdot \sin (3x + x) &= \sin 3x \\ \Rightarrow 4 [\sin x (\sin^2 3x - \sin^2 x)] &= 3 \sin x - 4 \sin^3 x \\ \Rightarrow 4 \sin x \sin^2 3x - 4 \sin^3 x &= 3 \sin x - 4 \sin^3 x \\ \Rightarrow 4 \sin x \sin^2 3x &= 3 \sin x \\ \Rightarrow \sin x (4 \sin^2 3x - 3) &= 0 \\ \Rightarrow \sin x = 0 \text{ or, } 4 \sin^2 3x - 3 &= 0 \\ \Rightarrow \sin x = 0 \text{ or, } \sin^2 3x &= \frac{3}{4} \end{aligned}$$

Now, $\sin x = 0 \Rightarrow x = n\pi, n \in \mathbb{Z}$

And, $\sin^2 3x = \frac{3}{4}$

$$\Rightarrow \sin^2 3x = \left(\frac{\sqrt{3}}{2}\right)^2$$

$$\Rightarrow \sin^2 3x = \sin^2 \frac{\pi}{3}$$

$$\Rightarrow 3x = m\pi \pm \frac{\pi}{3}, m \in \mathbb{Z}$$

$$\Rightarrow x = \frac{m\pi}{3} \pm \frac{\pi}{9}$$

Hence, $x = n\pi$ or, $x = m\pi \pm \frac{\pi}{3}$, where $m, n \in \mathbb{Z}$

11.4 TRIGONOMETRIC EQUATIONS OF THE FORM

$a \cos \theta + b \sin \theta = c$, where $a, b, c \in \mathbb{R}$ such that $|c| \leq \sqrt{a^2 + b^2}$

To solve this type of equations, we first reduce them in the form $\cos \theta = \cos \alpha$, or $\sin \theta = \sin \alpha$.

The following algorithm provides the method of solution.

ALGORITHM

STEP I Obtain the equation $a \cos \theta + b \sin \theta = c$.

STEP II Put $a = r \cos \alpha$ and $b = r \sin \alpha$, where $r = \sqrt{a^2 + b^2}$ and $\tan \alpha = b/a$ i.e. $\alpha = \tan^{-1}(b/a)$.

STEP III Using the substitution in step II, the equation reduces to

$$r \cos(\theta - \alpha) = c \Rightarrow \cos(\theta - \alpha) = \frac{c}{r} = \cos \beta \text{ (say).}$$

STEP IV Solve the equation obtained in step III by using the formulas discussed earlier.

ILLUSTRATIVE EXAMPLES

LEVEL-1

EXAMPLE 1 Solve: $\sqrt{3} \cos \theta + \sin \theta = \sqrt{2}$

[NCERT EXEMPLAR]

SOLUTION We have,

$$\sqrt{3} \cos \theta + \sin \theta = \sqrt{2}$$

...(i)

This is of the form $a \cos \theta + b \sin \theta = c$, where $a = \sqrt{3}$, $b = 1$ and $c = \sqrt{2}$.

Let $a = r \cos \alpha$ and $b = r \sin \alpha$. Then,

$$\sqrt{3} = r \cos \alpha \quad \text{and} \quad 1 = r \sin \alpha.$$

$$\Rightarrow r = \sqrt{a^2 + b^2} = \sqrt{(\sqrt{3})^2 + 1^2} = 2 \quad \text{and} \quad \tan \alpha = \frac{1}{\sqrt{3}} \Rightarrow \alpha = \frac{\pi}{6}$$

Substituting $a = \sqrt{3} = r \cos \alpha$ and $b = 1 = r \sin \alpha$ in the equation (i) it reduces to

$$r \cos \alpha \cos \theta + r \sin \alpha \sin \theta = \sqrt{2}$$

$$\Rightarrow r \cos(\theta - \alpha) = \sqrt{2}$$

$$\Rightarrow 2 \cos\left(\theta - \frac{\pi}{6}\right) = \sqrt{2}$$

$$\Rightarrow \cos\left(\theta - \frac{\pi}{6}\right) = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \cos\left(\theta - \frac{\pi}{6}\right) = \cos \frac{\pi}{4}$$

$$\Rightarrow \theta - \frac{\pi}{6} = 2n\pi \pm \frac{\pi}{4}, n \in \mathbb{Z}$$

$$\Rightarrow \theta = 2n\pi \pm \frac{\pi}{4} + \frac{\pi}{6}, n \in \mathbb{Z}$$

$$\Rightarrow \theta = 2n\pi + \frac{\pi}{4} + \frac{\pi}{6} \text{ or } \theta = 2n\pi - \frac{\pi}{4} + \frac{\pi}{6}$$

$$\Rightarrow \theta = 2n\pi + \frac{5\pi}{12} \text{ or } \theta = 2n\pi - \frac{\pi}{12}$$

$$\text{Hence, } \theta = 2n\pi + \frac{5\pi}{12} \text{ or } \theta = 2n\pi - \frac{\pi}{12}, \text{ where } n \in \mathbb{Z}$$

EXAMPLE 2 Solve: $\sqrt{2} \sec \theta + \tan \theta = 1$

SOLUTION We have,

$$\sqrt{2} \sec \theta + \tan \theta = 1$$

$$\Rightarrow \frac{\sqrt{2}}{\cos \theta} + \frac{\sin \theta}{\cos \theta} = 1$$

$$\Rightarrow \sqrt{2} + \sin \theta = \cos \theta$$

$$\Rightarrow \cos \theta - \sin \theta = \sqrt{2}$$

...(i)

This is of the form, $a \cos \theta - b \sin \theta = c$, where $a = 1$, $b = 1$ and $c = \sqrt{2}$

Let $a = r \cos \alpha$, and $b = r \sin \alpha$.

$$\Rightarrow 1 = r \cos \alpha \quad \text{and} \quad 1 = r \sin \alpha.$$

$$\Rightarrow r = \sqrt{a^2 + b^2} = \sqrt{1^2 + 1^2} = \sqrt{2} \text{ and, } \tan \alpha = \frac{r \sin \alpha}{r \cos \alpha} = 1$$

$$\Rightarrow r = \sqrt{2} \text{ and, } \alpha = \frac{\pi}{4}$$

Substituting $a = 1 = r \cos \alpha$ and $b = 1 = r \sin \alpha$ in (i), we get

$$r \cos \theta \cos \alpha - r \sin \theta \sin \alpha = \sqrt{2}$$

$$\Rightarrow r \cos(\theta + \alpha) = \sqrt{2}$$

$$\Rightarrow \cos\left(\theta + \frac{\pi}{4}\right) = 1$$

$$\Rightarrow \cos\left(\theta + \frac{\pi}{4}\right) = \cos 0^\circ$$

$$\Rightarrow \theta + \frac{\pi}{4} = 2n\pi \pm 0, n \in \mathbb{Z} \Rightarrow \theta = 2n\pi - \frac{\pi}{4}, n \in \mathbb{Z}$$

EXAMPLE 3 Solve: $\cot \theta + \operatorname{cosec} \theta = \sqrt{3}$

SOLUTION We have,

$$\cot \theta + \operatorname{cosec} \theta = \sqrt{3}$$

$$\Rightarrow \frac{\cos \theta}{\sin \theta} + \frac{1}{\sin \theta} = \sqrt{3}$$

$$\Rightarrow \cos \theta + 1 = \sqrt{3} \sin \theta$$

$$\Rightarrow \sqrt{3} \sin \theta - \cos \theta = 1$$

...(i)

This is of the form $a \sin \theta + b \cos \theta = c$, where $a = \sqrt{3}, b = -1$ and $c = 1$.

$$\therefore \sqrt{3} = r \sin \alpha \text{ and } 1 = r \cos \alpha$$

$$\Rightarrow r = \sqrt{a^2 + b^2} = \sqrt{3+1} = 2 \text{ and, } \tan \alpha = \frac{\sqrt{3}}{1} = \sqrt{3} \Rightarrow r = 2 \text{ and } \alpha = \pi/3$$

Substituting $a = \sqrt{3} = r \sin \alpha$ and $b = 1 = r \cos \alpha$ in (i), we get

$$r \sin \alpha \sin \theta - r \cos \alpha \cos \theta = 1$$

$$\Rightarrow -r \cos(\theta + \alpha) = 1$$

$$\Rightarrow -2 \cos\left(\theta + \frac{\pi}{3}\right) = 1$$

$$\Rightarrow \cos\left(\theta + \frac{\pi}{3}\right) = -\frac{1}{2}$$

$$\Rightarrow \cos\left(\theta + \frac{\pi}{3}\right) = \cos \frac{2\pi}{3}$$

$$\Rightarrow \theta + \frac{\pi}{3} = 2n\pi \pm \frac{2\pi}{3}, n \in \mathbb{Z}$$

$$\Rightarrow \theta = 2n\pi \pm \frac{2\pi}{3} - \frac{\pi}{3}, n \in \mathbb{Z}$$

$$\Rightarrow \theta = 2n\pi + \frac{\pi}{3}, n \in \mathbb{Z} \text{ or, } \theta = 2n\pi - \pi = (2n-1)\pi, n \in \mathbb{Z}$$

But, θ cannot be equal to $(2n-1)\pi$ as it makes $\sin \theta = 0$.

$$\text{Hence, } \theta = 2n\pi + \frac{\pi}{3}, n \in \mathbb{Z}$$

EXERCISE 11.1

LEVEL-1

1. Find the general solutions of the following equations:

$$\begin{array}{lll} \text{(i)} \sin \theta = \frac{1}{2} & \text{(ii)} \cos \theta = -\frac{\sqrt{3}}{2} & \text{(iii)} \operatorname{cosec} \theta = -\sqrt{2} \\ \text{(iv)} \sec \theta = \sqrt{2} & \text{(v)} \tan \theta = -\frac{1}{\sqrt{3}} & \text{(vi)} \sqrt{3} \sec \theta = 2 \end{array}$$

2. Find the general solutions of the following equations:

$$\begin{array}{lll} \text{(i)} \sin 2\theta = \frac{\sqrt{3}}{2} & \text{(ii)} \cos 3\theta = \frac{1}{2} & \text{(iii)} \sin 9\theta = \sin \theta \\ \text{(iv)} \sin 2\theta = \cos 3\theta & \text{(v)} \tan \theta + \cot 2\theta = 0 & \text{(vi)} \tan 3\theta = \cot \theta \\ \text{(vii)} \tan 2\theta \tan \theta = 1 & \text{(viii)} \tan m\theta + \cot n\theta = 0 & \text{(ix)} \tan p\theta = \cot q\theta \\ \text{(x)} \sin 2\theta + \cos \theta = 0 & \text{(xi)} \sin \theta = \tan \theta & \text{(xii)} \sin 3\theta + \cos 2\theta = 0 \end{array}$$

3. Solve the following equations:

$$\begin{array}{ll} \text{(i)} \sin^2 \theta - \cos \theta = \frac{1}{4} & \text{(ii)} 2 \cos^2 \theta - 5 \cos \theta + 2 = 0 \\ \text{(iii)} 2 \sin^2 x + \sqrt{3} \cos x + 1 = 0 & \text{(iv)} 4 \sin^2 \theta - 8 \cos \theta + 1 = 0 \\ \text{(v)} \tan^2 x + (1 - \sqrt{3}) \tan x - \sqrt{3} = 0 & \\ \text{(vi)} 3 \cos^2 \theta - 2\sqrt{3} \sin \theta \cos \theta - 3 \sin^2 \theta = 0 & \text{(vii)} \cos 4\theta = \cos 2\theta \end{array}$$

4. Solve the following equations:

$$\begin{array}{ll} \text{(i)} \cos \theta + \cos 2\theta + \cos 3\theta = 0 & \text{(ii)} \cos \theta + \cos 3\theta - \cos 2\theta = 0 \quad [\text{NCERT}] \\ \text{(iii)} \sin \theta + \sin 5\theta = \sin 3\theta & \text{(iv)} \cos \theta \cos 2\theta \cos 3\theta = \frac{1}{4} \\ \text{(v)} \cos \theta + \sin \theta = \cos 2\theta + \sin 2\theta & \text{(vi)} \sin \theta + \sin 2\theta + \sin 3\theta = 0 \\ \text{(vii)} \sin \theta + \sin 2\theta + \sin 3\theta + \sin 4\theta = 0 & \\ \text{(viii)} \sin 3\theta - \sin \theta = 4 \cos^2 \theta - 2 & \text{(ix)} \sin 2\theta - \sin 4\theta + \sin 6\theta = 0 \quad [\text{NCERT}] \end{array}$$

5. Solve the following equations:

$$\begin{array}{ll} \text{(i)} \tan \theta + \tan 2\theta + \tan 3\theta = 0 & \text{(ii)} \tan \theta + \tan 2\theta = \tan 3\theta \\ \text{(iii)} \tan 3\theta + \tan \theta = 2 \tan 2\theta & \end{array}$$

6. Solve the following equations:

$$\begin{array}{ll} \text{(i)} \sin \theta + \cos \theta = \sqrt{2} & \text{(ii)} \sqrt{3} \cos \theta + \sin \theta = 1 \\ \text{(iii)} \sin \theta + \cos \theta = 1 & \text{(iv)} \operatorname{cosec} \theta = 1 + \cot \theta \\ \text{(v)} (\sqrt{3} - 1) \cos \theta + (\sqrt{3} + 1) \sin \theta = 2 & \quad [\text{NCERT EXEMPLAR}] \end{array}$$

7. Solve the following equations:

$$\begin{array}{ll} \text{(i)} \cot \theta + \tan \theta = 2 & [\text{NCERT EXEMPLAR}] \\ \text{(ii)} 2 \sin^2 \theta = 3 \cos \theta, 0 \leq \theta \leq 2\pi & [\text{NCERT EXEMPLAR}] \\ \text{(iii)} \sec \theta \cos 5\theta + 1 = 0, 0 < \theta < \frac{\pi}{2} & [\text{NCERT EXEMPLAR}] \\ \text{(iv)} 5 \cos^2 \theta + 7 \sin^2 \theta - 6 = 0 & [\text{NCERT EXEMPLAR}] \\ \text{(v)} \sin x - 3 \sin 2x + \sin 3x = \cos x - 3 \cos 2x + \cos 3x & [\text{NCERT EXEMPLAR}] \end{array}$$

ANSWERS

$$\begin{array}{ll} 1. \text{ (i)} \theta = n\pi + (-1)^n \frac{\pi}{6}, n \in \mathbb{Z} & \text{(ii)} \theta = 2n\pi \pm \frac{7\pi}{6}, n \in \mathbb{Z} \\ \text{(iii)} \theta = n\pi + (-1)^{n+1} \frac{\pi}{4}, n \in \mathbb{Z} & \text{(iv)} \theta = 2n\pi \pm \frac{\pi}{4}, n \in \mathbb{Z} \end{array}$$

- (v) $\theta = n\pi - \frac{\pi}{6}, n \in \mathbb{Z}$ (vi) $\theta = 2n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$
2. (i) $\theta = \frac{n\pi}{2} + (-1)^n \frac{\pi}{6}, n \in \mathbb{Z}$ (ii) $\theta = \frac{2n\pi}{3} \pm \frac{\pi}{9}, n \in \mathbb{Z}$
- (iii) $\theta = \frac{r\pi}{4}$ or $\theta = (2r+1)\frac{\pi}{10}, \text{ where } r \in \mathbb{Z}$
- (iv) $\theta = (4n+1)\frac{\pi}{10}$ or $\theta = (4n-1)\frac{\pi}{10}, \text{ where } n \in \mathbb{Z}$
- (v) $\theta = n\pi - \frac{\pi}{2}, n \in \mathbb{Z}$ (vi) $\theta = \frac{n\pi}{4} + \frac{\pi}{8}, n \in \mathbb{Z}$
- (vii) $\theta = \frac{n\pi}{3} + \frac{\pi}{6}, n \in \mathbb{Z}$ (viii) $\theta = \frac{(2r+1)\pi}{m-n}, r \in \mathbb{Z}$
- (ix) $\theta = \left(\frac{2n+1}{p+q}\right)\frac{\pi}{2}, n \in \mathbb{Z}$
- (x) $\theta = (4n-1)\frac{\pi}{2}$ or $\theta = (4m-1)\frac{\pi}{6}, \text{ where } m, n \in \mathbb{Z}$
- (xi) $\theta = m\pi$ or $\theta = 2n\pi, \text{ where } m, n \in \mathbb{Z}$
- (xii) $\theta = (4n-1)\frac{\pi}{10}$ or $\theta = (4m-1)\frac{\pi}{2}, m, n \in \mathbb{Z}$
3. (i) $\theta = 2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$ (ii) $\theta = 2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$
- (iii) $\theta = 2n\pi \pm \frac{5\pi}{6}, n \in \mathbb{Z}$ (iv) $\theta = 2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$
- (v) $\theta = n\pi - \frac{\pi}{4}$ or $\theta = m\pi + \frac{\pi}{3}, \text{ where } m, n \in \mathbb{Z}$
- (vi) $\theta = n\pi - \frac{\pi}{3}$ or $\theta = m\pi + \frac{\pi}{6}, \text{ where } m, n \in \mathbb{Z}$
- (vii) $x = n\pi, x = \frac{n\pi}{3}, n \in \mathbb{Z}$
4. (i) $\theta = (2n+1)\frac{\pi}{4}$ or $\theta = 2m\pi \pm \frac{2\pi}{3}, \text{ where } m, n \in \mathbb{Z}$
- (ii) $\theta = (2n+1)\frac{\pi}{4}$ or $\theta = 2m\pi \pm \frac{\pi}{3}, \text{ where } m, n \in \mathbb{Z}$
- (iii) $\theta = \frac{n\pi}{3}$ or $\theta = m\pi \pm \frac{\pi}{6}, \text{ where } m, n \in \mathbb{Z}$
- (iv) $\theta = 2n + 1\frac{\pi}{8}$ or $\theta = m\pi \pm \frac{\pi}{3}, \text{ where } m, n \in \mathbb{Z}$
- (v) $\theta = \frac{(2n\pi)}{3} + \frac{\pi}{6}$ or $\theta = 2m\pi, \text{ where } m, n \in \mathbb{Z}$
- (vi) $\theta = \frac{n\pi}{2}$ or $\theta = 2n\pi \pm \frac{2\pi}{3}, \text{ where } m, n \in \mathbb{Z}$
- (vii) $\theta = n\pi + \frac{\pi}{2}, \theta = (2m+1)\pi, \theta = \frac{2r\pi}{5}, \text{ where } m, n, r \in \mathbb{Z}$
- (viii) $\theta = n\pi + (-1)^n \frac{\pi}{2}$ or $\theta = (2m+1)\frac{\pi}{4}, \text{ where } m, n \in \mathbb{Z}$
- (ix) $\theta = \frac{n\pi}{4}, \theta = n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$
5. (i) $\theta = \frac{m\pi}{3}$ or $\theta = n\pi \pm \alpha, \text{ where } \alpha = \tan^{-1} \frac{1}{\sqrt{2}} \text{ and } m, n \in \mathbb{Z}$
- (ii) $\theta = m\pi$ or $\theta = \frac{n\pi}{3}, \text{ where } m, n \in \mathbb{Z}$

(iii) $\theta = n\pi$, where $n \in \mathbb{Z}$

6. (i) $\theta = (8n + 1) \frac{\pi}{4}$, $n \in \mathbb{Z}$

(ii) $\theta = (4n + 1) \frac{\pi}{2}$ or $\theta = (12m - 1) \frac{\pi}{6}$, where $m, n \in \mathbb{Z}$

(iii) $\theta = 2n\pi$ or $\theta = 2m\pi + \frac{\pi}{2}$, where $m, n \in \mathbb{Z}$

(iv) $\theta = 2m\pi + \frac{\pi}{2}$, where $m, n \in \mathbb{Z}$

(v) $\theta = 2n\pi + \frac{\pi}{3}$ or, $\theta = 2n\pi - \frac{\pi}{6}$, $n \in \mathbb{Z}$

7. (i) $\theta = 2n\pi \pm \frac{\pi}{3}$, $n \in \mathbb{Z}$

(ii) $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$

(iii) $\theta = \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{2}$

(iv) $\theta = n\pi \pm \frac{\pi}{4}$, $n \in \mathbb{Z}$

(v) $x = \frac{n\pi}{2} \pm \frac{\pi}{8}$, $n \in \mathbb{Z}$

HINTS TO NCERT & SELECTED PROBLEMS

4. (ii)

$$\begin{aligned} (2n+1) \frac{\pi}{2}, n \in \mathbb{Z} &\Rightarrow \theta = (2n+1) \frac{\pi}{4}, n \in \mathbb{Z} \\ &= \cos \frac{\pi}{3} \Rightarrow \theta = 2m\pi \pm \frac{\pi}{3}, m \in \mathbb{Z} \end{aligned}$$

(iv) $3\theta = \frac{\pi}{4}$

$\Rightarrow 2(\cos \theta) \cos 3\theta = 1$

$\Rightarrow 2(\cos 3\theta + \cos \theta) \cos 3\theta = 1$

$\Rightarrow 2\cos^2 3\theta + 2\cos 3\theta \cos \theta = 1$

$\Rightarrow 2\cos^2 3\theta + \cos 4\theta + \cos 2\theta = 1$

$\Rightarrow (2\cos^2 3\theta - 1) + \cos 4\theta + \cos 2\theta = 0$

$\Rightarrow \cos 6\theta + \cos 2\theta + \cos 4\theta = 0$

$\Rightarrow 2\cos 4\theta \cos 2\theta + \cos 4\theta = 0$

$\Rightarrow \cos 4\theta (2\cos 2\theta + 1) = 0$

$\Rightarrow \cos 4\theta = 0, 2\cos 2\theta + 1 = 0$

$\Rightarrow \cos 4\theta = 0, \cos 2\theta = \cos \frac{2\pi}{3}$

$\Rightarrow 4\theta = (2n+1) \frac{\pi}{2}, 2\theta = 2n\pi \pm \frac{2\pi}{3}, n \in \mathbb{Z}$

$\Rightarrow \theta = (2n+1) \frac{\pi}{8}, \theta = n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$

(ix) We have,

$\sin 2\theta - \sin 4\theta + \sin 6\theta = 0$

$\Rightarrow \sin 6\theta + \sin 2\theta - \sin 4\theta = 0$

$\Rightarrow 2\sin 4\theta \cos 2\theta - \sin 4\theta = 0$

$$\Rightarrow \sin 4\theta (2 \cos 2\theta - 1) = 0$$

$$\Rightarrow \sin 4\theta = 0 \text{ or } 2 \cos 2\theta - 1 = 0$$

$$\text{Now, } \sin 4\theta = 0 \Rightarrow 4\theta = n\pi, n \in \mathbb{Z} \Rightarrow \theta = \frac{n\pi}{4}, n \in \mathbb{Z}$$

$$\text{and, } 2 \cos 2\theta - 1 = 0$$

$$\Rightarrow \cos 2\theta = \frac{1}{2} \Rightarrow \cos 2\theta = \cos \frac{\pi}{3} \Rightarrow 2\theta = 2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z} \Rightarrow \theta = n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$$

$$5. \text{ (i) } \tan \theta + \tan 2\theta + \tan 3\theta = 0$$

$$\Rightarrow \tan \theta + \tan 2\theta + \tan (\theta + 2\theta) = 0$$

$$\Rightarrow \tan \theta + \tan 2\theta + \frac{\tan \theta + \tan 2\theta}{1 - \tan \theta \tan 2\theta} = 0$$

$$\Rightarrow (\tan \theta + \tan 2\theta) (2 - \tan \theta \tan 2\theta) = 0$$

$$\Rightarrow \tan \theta + \tan 2\theta = 0 \text{ or } \tan \theta \tan 2\theta = 2$$

$$\text{Now, } \tan \theta + \tan 2\theta = 0$$

$$\Rightarrow \tan \theta + \frac{2 \tan \theta}{1 - \tan^2 \theta} = 0$$

$$\Rightarrow \tan \theta (3 - \tan^2 \theta) = 0$$

$$\Rightarrow \tan \theta = 0 \text{ or } \tan^2 \theta = 3$$

$$\Rightarrow \tan \theta = 0 \text{ or } \tan^2 \theta = \tan^2 \frac{\pi}{3}$$

$$\Rightarrow \theta = n\pi, \theta = n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$$

$$\text{and, } \tan \theta \tan 2\theta = 2$$

$$\Rightarrow \tan \theta \times \frac{2 \tan \theta}{1 - \tan^2 \theta} = 2$$

$$\Rightarrow \tan^2 \theta = 1 - \tan^2 \theta$$

$$\Rightarrow 2 \tan^2 \theta = 1$$

$$\Rightarrow \tan^2 \theta = \left(\frac{1}{\sqrt{2}}\right)^2 = \tan^2 \alpha \text{ (say), where } \tan \alpha = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \theta = n\pi \pm \alpha, \text{ where } \tan \alpha = \frac{1}{\sqrt{2}}$$

$$\text{(iii) } \tan 3\theta - \tan 2\theta = \tan 2\theta - \tan \theta$$

$$\Rightarrow \frac{\sin (3\theta - 2\theta)}{\cos 3\theta \cos 2\theta} = \frac{\sin (2\theta - \theta)}{\cos 2\theta \cos \theta}$$

$$\Rightarrow \frac{\sin \theta}{\cos 3\theta \cos 2\theta} = \frac{\sin \theta}{\cos 2\theta \cos \theta}$$

$$\Rightarrow \sin \theta \cos 2\theta (\cos 3\theta - \cos \theta) = 0$$

$$\Rightarrow -2 \sin \theta \cos 2\theta \sin 2\theta \sin \theta = 0$$

$$\Rightarrow 2 \sin^2 \theta \sin 2\theta = 0$$

$$[\because \cos 2\theta \neq 0]$$

$$\Rightarrow \sin \theta = 0 \text{ or } \sin 2\theta = 0 \Rightarrow \theta = n\pi \text{ or } 2\theta = m\pi \Rightarrow \theta = n\pi \text{ or } \theta = \frac{m\pi}{2}, \text{ where } n, m \in \mathbb{Z}$$

VERY SHORT ANSWER QUESTIONS (VSAQs)

Answer each of the following questions in one word or one sentence or as per exact requirement of the question:

- Write the number of solutions of the equation $\tan x + \sec x = 2 \cos x$ in the interval $[0, 2\pi]$.
- Write the number of solutions of the equation $4 \sin x - 3 \cos x = 7$.
- Write the general solution of $\tan^2 2x = 1$.
- Write the set of values of a for which the equation $\sqrt{3} \sin x - \cos x = a$ has no solution.
- If $\cos x = k$ has exactly one solution in $[0, 2\pi]$, then write the value(s) of k .
- Write the number of points of intersection of the curves $2y = 1$ and $y = \cos x$, $0 \leq x \leq 2\pi$.
- Write the values of x in $[0, \pi]$ for which $\sin 2x$, $\frac{1}{2}$ and $\cos 2x$ are in A.P.
- Write the number of points of intersection of the curves $2y = -1$ and $y = \operatorname{cosec} x$.
- Write the solution set of the equation $(2 \cos \theta + 1)(4 \cos \theta + 5) = 0$ in the interval $[0, 2\pi]$.
- Write the number of values of θ in $[0, 2\pi]$ that satisfy the equation $\sin^2 \theta - \cos \theta = \frac{1}{4}$.
- If $3 \tan(\theta - 15^\circ) = \tan(\theta + 15^\circ)$, $0 < \theta < 90^\circ$, find θ .
- If $2 \sin^2 \theta = 3 \cos \theta$, where $0 \leq \theta \leq 2\pi$, then find the value of θ .
- If $\sec x \cos 5x + 1 = 0$, where $0 < x \leq \frac{\pi}{2}$, find the value of x .

ANSWERS

- | | | | |
|--|-------|---|--|
| 1. 2 | 2. 0 | 3. $\frac{n\pi}{2} + \frac{\pi}{8}, n \in \mathbb{Z}$ | 4. $a \in (-\infty, -2) \cup (2, \infty)$ |
| 5. -1 | 6. 2 | 7. $0, \frac{\pi}{4}, \pi$ | 8. 0 |
| 9. $\frac{2\pi}{3}, \frac{4\pi}{3}$ | 10. 2 | 11. $\frac{\pi}{4}$ | 12. $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$ |
| 13. $\theta = \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{2}$ | | | |

MULTIPLE CHOICES QUESTIONS (MCQs)

Mark the correct alternative in each of the following:

- The smallest value of θ satisfying the equation $\sqrt{3}(\cot \theta + \tan \theta) = 4$ is
 (a) $2\pi/3$ (b) $\pi/3$ (c) $\pi/6$ (d) $\pi/12$
- If $\cos \theta + \sqrt{3} \sin \theta = 2$, then $\theta =$
 (a) $\pi/3$ (b) $2\pi/3$ (c) $4\pi/3$ (d) $5\pi/3$
- If $\tan p\theta - \tan q\theta = 0$, then the values of θ form a series in
 (a) AP (b) GP (c) HP (d) none of these
- If a is any real number, the number of roots of $\cot x - \tan x = a$ in the first quadrant is (are).
 (a) 2 (b) 0 (c) 1 (d) none of these
- The general solution of the equation $7 \cos^2 \theta + 3 \sin^2 \theta = 4$ is
 (a) $\theta = 2n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$ (b) $\theta = 2n\pi \pm \frac{2\pi}{3}, n \in \mathbb{Z}$

- (c) $\theta = n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$ (d) none of these
6. A solution of the equation $\cos^2 \theta + \sin \theta + 1 = 0$, lies in the interval
 (a) $(-\pi/4, \pi/4)$ (b) $(\pi/4, 3\pi/4)$
 (c) $(3\pi/4, 5\pi/4)$ (d) $(5\pi/4, 7\pi/4)$
7. The number of solution in $[0, \pi/2]$ of the equation $\cos 3x \tan 5x = \sin 7x$ is
 (a) 5 (b) 7 (c) 6 (d) none of these
8. The general value of x satisfying the equation $\sqrt{3} \sin x + \cos x = \sqrt{3}$ is given by
 (a) $x = n\pi + (-1)^n \frac{\pi}{4} + \frac{\pi}{3}, n \in \mathbb{Z}$ (b) $x = n\pi + (-1)^n \frac{\pi}{3} - \frac{\pi}{6}, n \in \mathbb{Z}$
 (c) $x = n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$ (d) $x = n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$
9. The smallest positive angle which satisfies the equation
 $2 \sin^2 \theta + \sqrt{3} \cos \theta + 1 = 0$ is
 (a) $\frac{5\pi}{6}$ (b) $\frac{2\pi}{3}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$
10. If $4 \sin^2 \theta = 1$, then the values of θ are
 (a) $2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$ (b) $n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$ (c) $n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$ (d) $2n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$
11. If $\cot \theta - \tan \theta = \sec \theta$, then, θ is equal to
 (a) $2n\pi + \frac{3\pi}{2}, n \in \mathbb{Z}$ (b) $n\pi + (-1)^n \frac{\pi}{6}, n \in \mathbb{Z}$
 (c) $n\pi + \frac{\pi}{2}, n \in \mathbb{Z}$ (d) none of these.
12. A value of θ satisfying $\cos \theta + \sqrt{3} \sin \theta = 2$ is
 (a) $\frac{5\pi}{3}$ (b) $\frac{4\pi}{3}$ (c) $\frac{2\pi}{3}$ (d) $\frac{\pi}{3}$
13. In $(0, \pi)$, the number of solutions of the equation
 $\tan \theta + \tan 2\theta + \tan 3\theta = \tan \theta \tan 2\theta \tan 3\theta$ is
 (a) 7 (b) 5 (c) 4 (d) 2.
14. The number of values of θ in $[0, 2\pi]$ that satisfy the equation $\sin^2 \theta - \cos \theta = \frac{1}{4}$
 (a) 1 (b) 2 (c) 3 (d) 4
15. If $e^{\sin x} - e^{-\sin x} - 4 = 0$, then $x =$
 (a) 0 (b) $\sin^{-1}\{\log_e(2 - \sqrt{5})\}$
 (c) 1 (d) none of these
16. The equation $3 \cos x + 4 \sin x = 6$ has solution.
 (a) finite (b) infinite (c) one (d) no
17. If $\sqrt{3} \cos \theta + \sin \theta = \sqrt{2}$, then general value of θ is
 (a) $n\pi + (-1)^n \frac{\pi}{4}, n \in \mathbb{Z}$ (b) $(-1)^n \frac{\pi}{4} - \frac{\pi}{3}, n \in \mathbb{Z}$

(c) $n\pi + \frac{\pi}{4} - \frac{\pi}{3}, n \in \mathbb{Z}$

(d) $n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{3}, n \in \mathbb{Z}$

18. General solution of $\tan 5\theta = \cot 2\theta$ is

(a) $\frac{n\pi}{7} + \frac{\pi}{2}, n \in \mathbb{Z}$

(b) $\theta = \frac{n\pi}{7} + \frac{\pi}{3}, n \in \mathbb{Z}$

(c) $\theta = \frac{n\pi}{7} + \frac{\pi}{14}, n \in \mathbb{Z}$

(d) $\theta = \frac{n\pi}{7} - \frac{\pi}{14}, n \in \mathbb{Z}$

19. The solution of the equation $\cos^2 \theta + \sin \theta + 1 = 0$ lies in the interval

(a) $(-\pi/4, \pi/4)$ (b) $(\pi/4, 3\pi/4)$ (c) $(3\pi/4, 5\pi/4)$ (d) $(5\pi/4, 7\pi/4)$

20. If $\cos \theta = -\frac{1}{2}$ and $0 < \theta < 360^\circ$, then the solutions are

(a) $\theta = 60^\circ, 240^\circ$

(b) $\theta = 120^\circ, 240^\circ$

(c) $\theta = 120^\circ, 210^\circ$

(d) $\theta = 120^\circ, 300^\circ$

21. The number of values of x in the interval $[0, 5\pi]$ satisfying the equation $3\sin^2 x - 7\sin x + 2 = 0$ is

(a) 0

(b) 5

(c) 6

(d) 10

ANSWERS

1. (c) 2. (a) 3. (a) 4. (c) 5. (a) 6. (d) 7. (c) 8. (b)
 9. (a) 10. (c) 11. (b) 12. (d) 13. (d) 14. (b) 15. (d) 16. (d)
 17. (d) 18. (c) 19. (d) 20. (b) 21. (c)

SUMMARY

1. An equation containing trigonometric functions of unknown angles is known as a trigonometric equation.
2. A solution of a trigonometric equation is the value of the unknown angle that satisfies the equation.
3. Following are the general solutions of trigonometric equations in standard forms:

*Trigonometric equation**General solution*

(i) $\sin \theta = 0$

$\theta = n\pi, n \in \mathbb{Z}$

(ii) $\cos \theta = 0$

$\theta = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$

(iii) $\tan \theta = 0$

$\theta = n\pi, n \in \mathbb{Z}$

(iv) $\sin \theta = \sin \alpha$

$\theta = n\pi + (-1)^n \alpha, n \in \mathbb{Z}$

(v) $\cos \theta = \cos \alpha$

$\theta = 2n\pi \pm \alpha, n \in \mathbb{Z}$

(vi) $\tan \theta = \tan \alpha$

$\theta = n\pi + \alpha, n \in \mathbb{Z}$

$$\left. \begin{aligned} \sin^2 \theta &= \sin^2 \alpha \\ \cos^2 \theta &= \cos^2 \alpha \\ \tan^2 \theta &= \tan^2 \alpha \end{aligned} \right\}$$

$\theta = n\pi \pm \alpha, n \in \mathbb{Z}$

4. The equation $a \cos \theta + b \sin \theta = c$ is solvable for $|c| \leq \sqrt{a^2 + b^2}$.

CHAPTER 12

MATHEMATICAL INDUCTION

12.1 STATEMENTS

A sentence or description which can be judged to be true or false is called a statement.

Following are some examples of statements:

EXAMPLE 1 2 divides 6.

EXAMPLE 2 Jaipur is the capital of Rajasthan.

EXAMPLE 3 There are 5 days in a week.

EXAMPLE 4 $(x + 1)$ is a factor of $x^2 - 3x + 2$.

EXAMPLE 5 $A \cup B = B \cup A$.

Clearly, statements in Examples 1, 2 and 5 are true statements whereas statements in Examples 3 and 4 are false.

MATHEMATICAL STATEMENTS Statements involving mathematical relations are known as the mathematical statements.

Clearly, statements in examples 1, 4 and 5 are mathematical statements. In this chapter, we shall be mainly discussing mathematical statements concerning natural numbers. We shall be using notations $P(n)$ or $P_1(n)$ or $P_2(n)$ etc. to denote such statements.

EXAMPLE 1 Let $P(n)$ be the statement " $10n + 3$ is prime". Then, $P(2)$ is the statement " $10 \times 2 + 3$ is prime" i.e. "23 is prime".

Clearly, $P(2)$ is true.

$P(3)$ is the statement " $10 \times 3 + 3$ is prime" i.e. "33 is prime".

Clearly $P(3)$ is not true.

EXAMPLE 2 If $P(n)$ is the statement " $n^3 + n$ is divisible by 3", is the statement $P(3)$ true? Is the statement $P(4)$ true?

SOLUTION $P(3)$ is the statement " $3^3 + 3 = 30$ is divisible by 3".

Clearly, it is true.

$P(4)$ is the statement " $4^3 + 4 = 68$ is divisible by 3".

Clearly, it is not true.

EXAMPLE 3 If $P(n)$ is the statement " $n(n+1)(n+2)$ is divisible by 12", prove that the statements $P(3)$ and $P(4)$ are true, but that $P(5)$ is not true.

SOLUTION $P(3)$ is the statement " $3(3+1)(3+2) = 60$ is divisible by 12".

It is true.

$P(4)$ is the statement " $4(4+1)(4+2) = 120$ is divisible by 12".

It is also true.

$P(5)$ is the statement " $5(5+1)(5+2) = 210$ is divisible by 12".

Clearly it is not true.

EXAMPLE 4 Let $P(n)$ be the statement " 7 divides $(2^{3n} - 1)$ ". What is $P(n+1)$?

SOLUTION $P(n+1)$ is the statement " 7 divides $(2^{3(n+1)} - 1)$ ".

Clearly, $P(n+1)$ is obtained by replacing n by $(n+1)$ in $P(n)$.

EXAMPLE 5 If $P(n)$ is the statement " $n^2 > 100$ ", prove that whenever $P(r)$ is true, $P(r+1)$ is also true.

SOLUTION The statement $P(n)$ is " $n^2 > 100$ ". Let $P(r)$ be true. Then $r^2 > 100$.

We wish to prove that the statement $P(r+1)$ is true i.e. " $(r+1)^2 > 100$ ".

Now,

$$\begin{aligned} & P(r) \text{ is true} \\ \Rightarrow & r^2 > 100 \\ \Rightarrow & r^2 + 2r + 1 > 100 + 2r + 1 && \text{[Adding } (2r + 1) \text{ on both sides]} \\ \Rightarrow & (r+1)^2 > 100 + 2r + 1 \\ \Rightarrow & (r+1)^2 > 100 \\ \Rightarrow & P(r+1) \text{ is true} && [\because 100 + 2r + 1 > 100 \text{ for every natural number } r] \end{aligned}$$

Thus, whenever $P(r)$ is true, $P(r+1)$ is also true.

EXAMPLE 6 Let $P(n)$ be the statement " $3^n > n$ ". If $P(n)$ is true, prove that $P(n+1)$ is true.

SOLUTION We are given that $P(n)$ is true i.e. $3^n > n$, and we wish to prove that $P(n+1)$ is true i.e. $3^{n+1} > (n+1)$.

Now,

$$\begin{aligned} & P(n) \text{ is true} \\ \Rightarrow & 3^n > n \\ \Rightarrow & 3 \cdot 3^n > 3n && \text{[Multiplying both sides by 3]} \\ \Rightarrow & 3^{n+1} > n + 2n \\ \Rightarrow & 3^{n+1} > n + 1 && [\because 2n > 1 \text{ for every } n \in \mathbb{N} \Rightarrow 2n + n > n + 1 \text{ for every } n \in \mathbb{N}] \\ \Rightarrow & P(n+1) \text{ is true} \end{aligned}$$

EXAMPLE 7 If $P(n)$ is the statement " $2^{3n} - 1$ is an integral multiple of 7 ", and if $P(r)$ is true, prove that $P(r+1)$ is true.

SOLUTION Let $P(r)$ be true. Then, $2^{3r} - 1$ is an integral multiple of 7 .

We wish to prove that $P(r+1)$ is true i.e. $2^{3(r+1)} - 1$ is an integral multiple of 7 .

Now,

$$\begin{aligned} & P(r) \text{ is true} \\ \Rightarrow & 2^{3r} - 1 \text{ is an integral multiple of } 7 \\ \Rightarrow & 2^{3r} - 1 = 7\lambda, \text{ for some } \lambda \in \mathbb{N}. \\ \Rightarrow & 2^{3r} = 7\lambda + 1 && \dots(i) \\ \text{Now, } & 2^{3(r+1)} - 1 = 2^{3r} \times 2^3 - 1 = (7\lambda + 1) \times 8 - 1 && \text{[Using (i)]} \\ \Rightarrow & 2^{3(r+1)} - 1 = 56\lambda + 8 - 1 = 56\lambda + 7 = 7(8\lambda + 1) \\ \Rightarrow & 2^{3(r+1)} - 1 = 7\mu, \text{ where } \mu = 8\lambda + 1 \in \mathbb{N} \\ \Rightarrow & 2^{3(r+1)} - 1 \text{ is an integral multiple of } 7 \\ \Rightarrow & P(r+1) \text{ is true} \end{aligned}$$

EXERCISE 12.1

1. If $P(n)$ is the statement " $n(n+1)$ is even", then what is $P(3)$?
2. If $P(n)$ is the statement " $n^3 + n$ is divisible by 3", prove that $P(3)$ is true but $P(4)$ is not true.
3. If $P(n)$ is the statement " $2^n \geq 3n$ ", and if $P(r)$ is true, prove that $P(r+1)$ is true.
4. If $P(n)$ is the statement " $n^2 + n$ is even", and if $P(r)$ is true, then $P(r+1)$ is true.
5. Given an example of a statement $P(n)$ such that it is true for all $n \in N$.
6. If $P(n)$ is the statement " $n^2 - n + 41$ is prime", prove that $P(1)$, $P(2)$ and $P(3)$ are true.
Prove also that $P(41)$ is not true.
7. Give an example of a statement $P(n)$ which is true for all $n \geq 4$ but $P(1)$, $P(2)$ and $P(3)$ are not true. Justify your answer.

ANSWERS

1. $P(3) : 3(3+1)$ is even
5. $P(n) : 1 + 2 + \dots + n = \frac{n(n+1)}{2}$
7. $P(n) : 2n < n!$

HINTS TO SELECTED PROBLEMS

3. Let $P(r)$ be true. Then, $P(r)$ is true
 $\Rightarrow 2^r \geq 3r$
 $\Rightarrow 2 \cdot 2^r \geq 6r$
 $\Rightarrow 2^{r+1} \geq 3r + 3r$
 $\Rightarrow 2^{r+1} \geq 3r + 3$ $[\because 3r \geq 3 \Rightarrow 3r + 3r \geq 3r + 3]$
 $\Rightarrow 2^{r+1} \geq 3(r+1) \Rightarrow P(r+1)$ is true
5. See the statement in Q. No. 4

12.2 THE PRINCIPLES OF MATHEMATICAL INDUCTION

FIRST PRINCIPLE OF MATHEMATICAL INDUCTION

Let $P(n)$ be a statement involving the natural number n such that

(I) $P(1)$ is true i.e. $P(n)$ is true for $n = 1$.

and, (II) $P(m+1)$ is true, whenever $P(m)$ is true.

i.e. $P(m)$ is true $\Rightarrow P(m+1)$ is true.

Then, $P(n)$ is true for all natural numbers n .

SECOND PRINCIPLE OF MATHEMATICAL INDUCTION

Let $P(n)$ be a statement involving the natural number n such that

(I) $P(1)$ is true i.e. $P(n)$ is true for $n = 1$.

and, (II) $P(m+1)$ is true, whenever $P(n)$ is true for all n , where $1 \leq n \leq m$.

Then, $P(n)$ is true for all natural numbers.

ILLUSTRATIVE EXAMPLES

Type I PROBLEMS BASED UPON FIRST PRINCIPLE OF MATHEMATICAL INDUCTION

Recall that the first principle of mathematical induction consists of two parts. First we must show that the given statement $P(n)$ is true for $n = 1$. The second part has two steps. The first step is to assume that the statement $P(n)$ is true for some $m \in N$. The second step is to use this assumption to prove that the statement $P(n)$ is true for $n = m + 1$.

In order to prove that a statement is true for all natural numbers using first principle of mathematical induction, we may use the following algorithm:

ALGORITHM

STEP I Obtain $P(n)$ and understand its meaning.

STEP II Prove that the statement $P(1)$ is true i.e. $P(n)$ is true for $n = 1$.

STEP III Assume that the statement $P(n)$ is true for $n = m$ (say) i.e. $P(m)$ is true.

STEP IV Using assumption in step III prove that $P(m + 1)$ is true.

STEP V Combining the results of step II and step IV, conclude by the first principle of mathematical induction that $P(n)$ is true for all $n \in N$.

The following examples illustrate the above algorithm.

LEVEL-1

EXAMPLE 1 Prove by the principle of mathematical induction that for all $n \in N$, $n^2 + n$ is even natural number

SOLUTION Let $P(n)$ be the statement " $n^2 + n$ is even".

STEP I We have, $P(n) : n^2 + n$ is even

$$\therefore 1^2 + 1 = 2, \text{ which is even}$$

$$\therefore P(1) \text{ is true}$$

STEP II Let $P(m)$ be true. Then,

$$P(m) \text{ is true} \Rightarrow m^2 + m \text{ is even} \Rightarrow m^2 + m = 2\lambda \text{ for some } \lambda \in N \quad \dots(i)$$

Now, we shall show that $P(m + 1)$ is true. For this we have to show that $(m + 1)^2 + (m + 1)$ is an even natural number.

Now,

$$(m + 1)^2 + (m + 1) = (m^2 + 2m + 1) + (m + 1) = (m^2 + m) + (2m + 2)$$

$$\Rightarrow (m + 1)^2 + (m + 1) = m^2 + m + 2(m + 1) = 2\lambda + 2(m + 1) \quad [\text{Using (i)}]$$

$$\Rightarrow (m + 1)^2 + (m + 1) = 2(\lambda + m + 1) = 2\mu, \text{ where } \mu = \lambda + m + 1 \in N$$

$$\Rightarrow (m + 1)^2 + (m + 1) \text{ is an even natural number}$$

$$\Rightarrow P(m + 1) \text{ is true}$$

$$\text{Thus, } P(m) \text{ is true} \Rightarrow P(m + 1) \text{ is true}$$

Hence, by the principle of mathematical induction, $P(n)$ is true for all $n \in N$ i.e. $n^2 + n$ is even for all $n \in N$.

EXAMPLE 2 Prove by the principle of mathematical induction that $n(n + 1)(2n + 1)$ is divisible by 6 for all $n \in N$.

SOLUTION Let $P(n)$ be the statement " $n(n + 1)(2n + 1)$ is divisible by 6".

i.e. $P(n) : n(n + 1)(2n + 1)$ is divisible by 6

STEP I We have, $P(1) : 1(1 + 1)(2 + 1)$ is divisible by 6.

$$\therefore 1(1 + 1)(2 + 1) = 6 \text{ which is divisible by 6}$$

$$\therefore P(1) \text{ is true}$$

STEP II Let $P(m)$ be true. Then,

$$m(m + 1)(2m + 1) \text{ is divisible by 6}$$

$$\Rightarrow m(m + 1)(2m + 1) = 6\lambda, \text{ for some } \lambda \in N \quad \dots(i)$$

Now, we shall show that $P(m + 1)$ is true. For this we have to show that $(m + 1)(m + 1 + 1)(2(m + 1) + 1)$ is divisible by 6.